CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Floating Point Representation

Lecture No.- 1



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Recap of Previous Lecture



- **Topic** Control Unit
- **Topic** Hardwired Control Unit
- **Topic** Micro-Programmed Control Unit
- **Topic** RISC vs CISC

Topics to be Covered



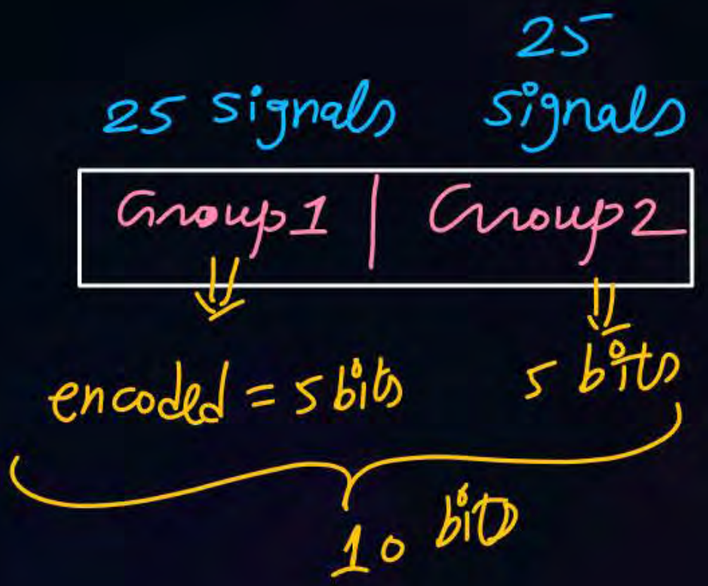
- Topic: RISC vs CISC
- Topic: Floating-Point Numbers
- Topic: Biased Exponent
- Topic: Number Range
- Topic: IEEE-754 Floating Point Representation

[NAT]

Ans = 10

#Q. A micro-programmed control unit is required to generate a total of 25 control signals. Assume that during any microinstruction at most 2 control signals are active. Minimum number of bits required in the control word to generate the required control signals will be?

vertical



assume 25 signals ⇒ S1, S2, ... , S25

Example:-

microprogrammed C.U.

16 inst^{ns} supported

For each inst^{n} ⇒ 8 microoperat^{ns} → no. of microinst^{ns} = 16 * 8

= 128 = 2^7

⇓

add. = 7 bits

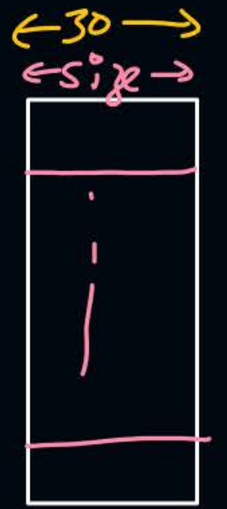
Microinst^{n} ⇒ 30 bits



No. of inputs in MUX = 8 ⇒ MUX select = 3 bits

Control mem. size = 3840 bits ?

↳ 2^7 * 30 bits = 3840 bits



no. of microinst^{ns} = 128

[NAT]

#Q. Consider a microprogrammed control unit which has to support 64 number of instructions. For each instruction execution control unit generates a sequence of 64 control words. Each microinstruction contains 3 fields: 122 control signals to support horizontal control unit, a MUX select field to select one of 7 inputs, and a next address field. The size of control memory needed is?

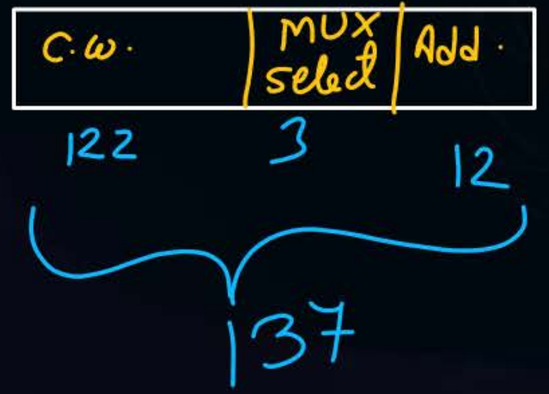
→ MUX select = 3 bits

no. of microinstns = 64 * 64

$= 2^{12}$

⇓

add = 12 bits



Control mem. size

$= 2^{12} * 137$ bits

$= 2^{10} * 2^{2} * 137$ bits

= 548 Kbits

Ans.

[NAT]

H.W.

#Q. Design of a vertical microprogrammed control unit requires to generate 50 signals. Out of first 43 those only 4 signals can be active at a time. And for remaining 7, anyone can be active anytime. The microinstruction of the control unit stores control signal information along with 3-bit mux select and 10-bits address field. The size of control memory required is?



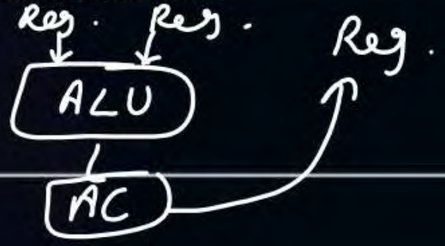
Topic : RISC vs CISC

S. No.	RISC (Reduced Instruction-Set Computer)	CISC (Complex Instruction-Set Computer)
1.	Less Number of Instructions Supported	More Number of Instructions
2.	Fixed Length Instructions	Variable Length Instructions
3.	Simple Instructions	Complex Instructions
4.	Simple and less number of addressing Modes	Complex and More number of addressing Modes
5.	Easy to implement using hardwired control unit	Difficult to implement using hardwired control unit



Topic : RISC vs CISC

S. No.	RISC (Reduced Instruction-Set Computer)	CISC (Complex Instruction-Set Computer)
6.	One Cycle per instruction	More than one cycle per instruction
7.	Register-to-Register arithmetic operation only	Register-to-Memory & Memory-to-Register arithmetic operations possible
8.	More Number of Registers	Less Number of Registers



[MCQ]

GATE- 2018 PYQ

#Q. Consider the following processor design characteristics.

I. Register-to register arithmetic operations only

II. Fixed-length instruction format

III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

A. I and II only

B. II and III only

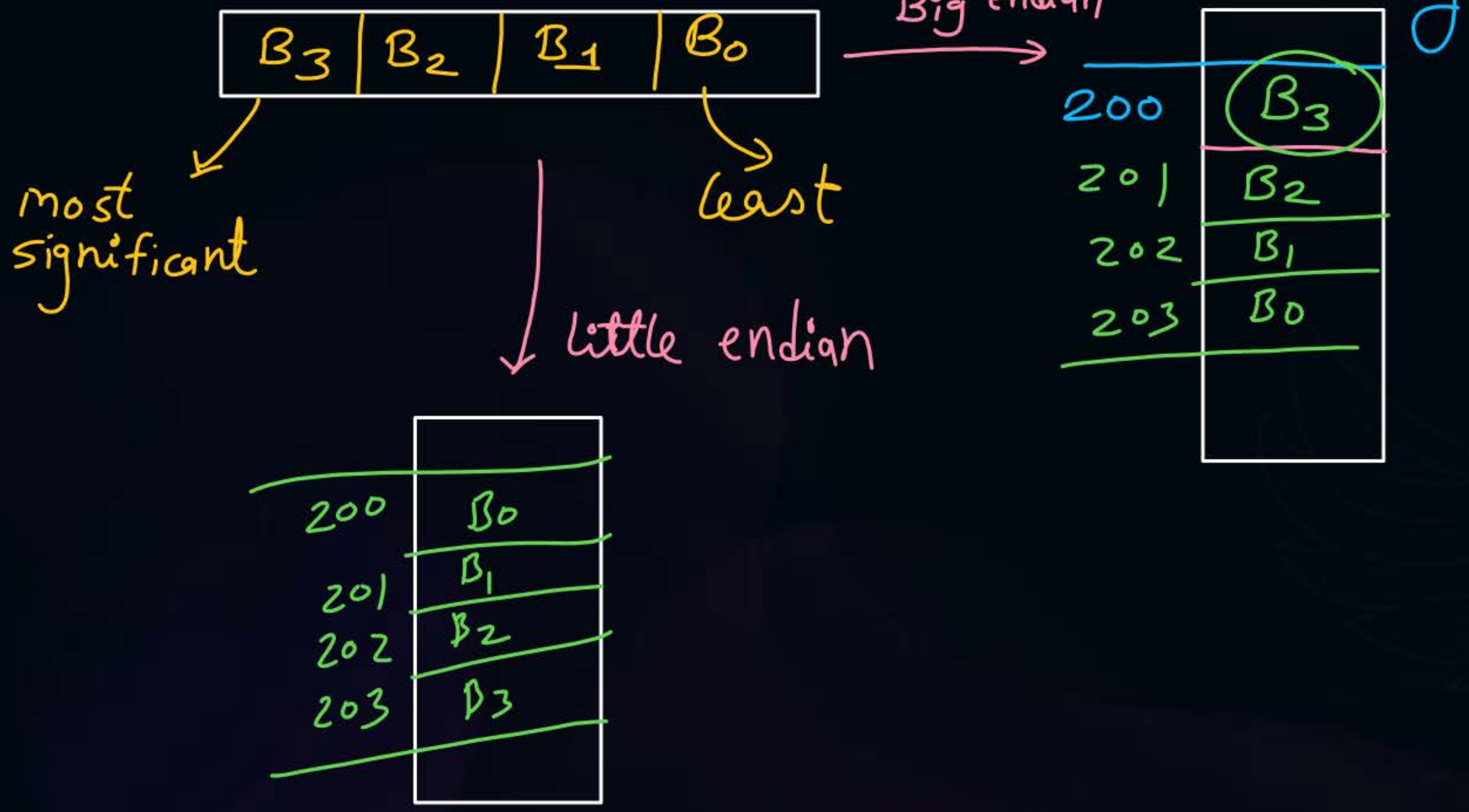
C. I and III only

D. I, II and III



Topic : Byte Ordering

Assume a content of 4 bytes ⇒ to be stored in a byte addressable mem.



Instⁿ cycle for variable length instⁿ :-

1. CPU fetches opcode only first
2. CPU decodes opcode, and gets the instⁿ length
   then now CPU fetches remaining part of instⁿ,
   and increments PC appropriately.

Floating Point Representation

⇓

motive => to be able to store a larger range of numbers with limited bits.



Topic : Floating-Point Numbers

- The number is represented in format:

Sign	Exponent	Mantissa
S	E	M

- Mantissa is signed normalized (implicit/explicit) fraction number
- Exponent is stored in biased form.

always 1 bit ⇒ S { 0 +ve, 1 −ve }

number of binary

⇓

format

⇓

$0.(1101) * 2^{(4)}$

(4) → exponent

(1101) → mantissa

Topic : Biased Exponent

$(0.101 * 2^6) + 0.110 * 2^5 \Rightarrow$ easier

vs

$0.101 * 2^{-2} + 0.1101 * 2^3$



← Signed →

Range ⇒ −8 to +7 → Transform → 0 to 15

← unsigned →

$E = e + bias$

- E : stored exponent
- e : original exponent

e	E	$E = e + 8$
−8	0	bias = 8
−7	1	
−6	2	
⋮	⋮	
0	8	
⋮	⋮	
+7	15	

→ excess − 8

if no. of bits used for representing $E = k$

then bias $= 2^{k-1}$

---

if bias $= 64$

$k = ?$ 7

$2^{k-1} = 2^{6}$

$k - 1 = 6$

$k = 7$



Topic : Mantissa

ex:-

10|.11

Explicit normalization $\Rightarrow$ $0.10111 * 2^{3}$

only 0 | should be 1

original exponent = 3

biased exponent = 3 + bias

mantissa = number after point

= 10111

Implicit normalization $\Rightarrow$ $1.0111 * 2^{2}$

should be 1

e = 2

E = 2 + bias

m = 0111



Topic : Value Formula

S	E	M

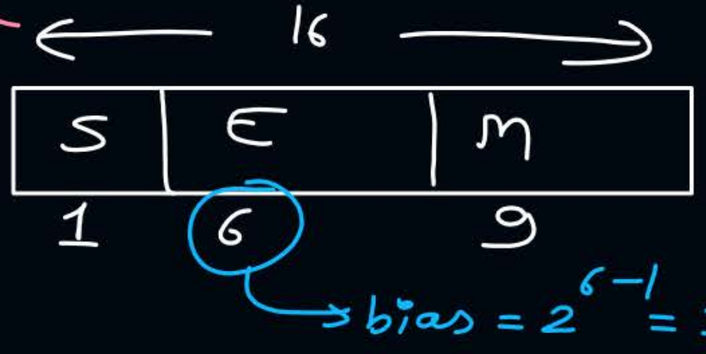
$S = 0 \quad (-1)^0 = 1 \Rightarrow +ve$

$S = 1 \quad (-1)^1 = -1 \Rightarrow -ve$

$$Value_{(explicit)} = (-1)^S * 0.M * 2^{E-bias}$$

$$Value_{(Implicit)} = (-1)^S * 1.M * 2^{E-bias}$$

Assume

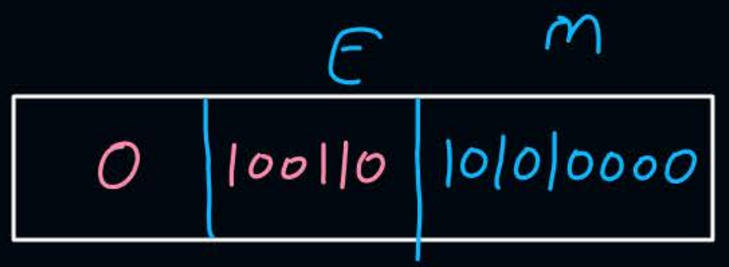


number = + 0.10101 * 2^{6}

6 → e = 6

E = 6 + 32 = $(38)_{10}$ = $(100110)_2$

m = 10101



[MCQ]

#Q. A certain well-known computer family represents the exponents of its floating-point numbers as "excess-64" integers; i.e., a typical exponent $e_6e_5e_4e_3e_2e_1e_0$ represents the number:

A $e = -64 + \sum_{i=0}^{6} 2iei$

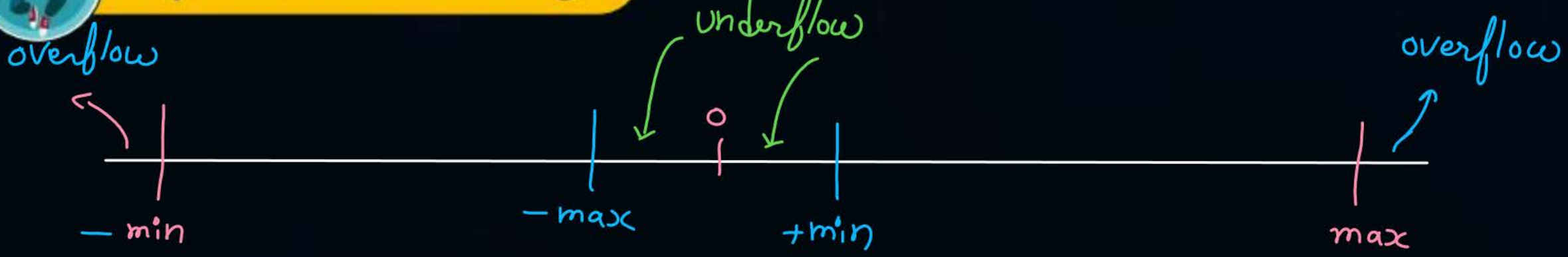
B $e = -64 + \sum_{i=0}^{6} 2e_i$

C $e = 64 - \sum_{i=0}^{6} 2iei$

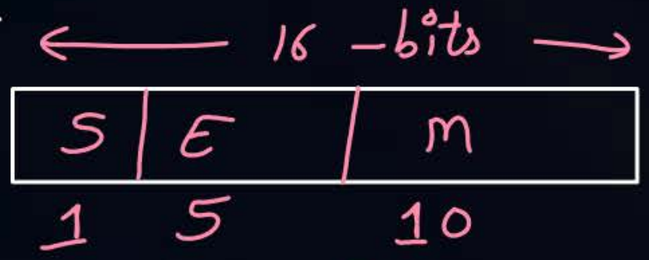
D $e = 64 - \sum_{i=0}^{6} 2e_i$



Topic : Number Range



ex:-



bias $= 2^{5-1} = 2^4 = 16$

unsigned value range = 0 to 31 ⇒ Range of e −16 to +15

(range of E)

assume number :-

$0.\underbrace{000000000000000000}_{17\ 0's}101$

⇓

explicit normalizatn

⇓

$0.101 * 2^{-17}$ → out of range

↓

Can't be stored

⇓

Underflow



Topic : Disadvantages of Conventional Representation

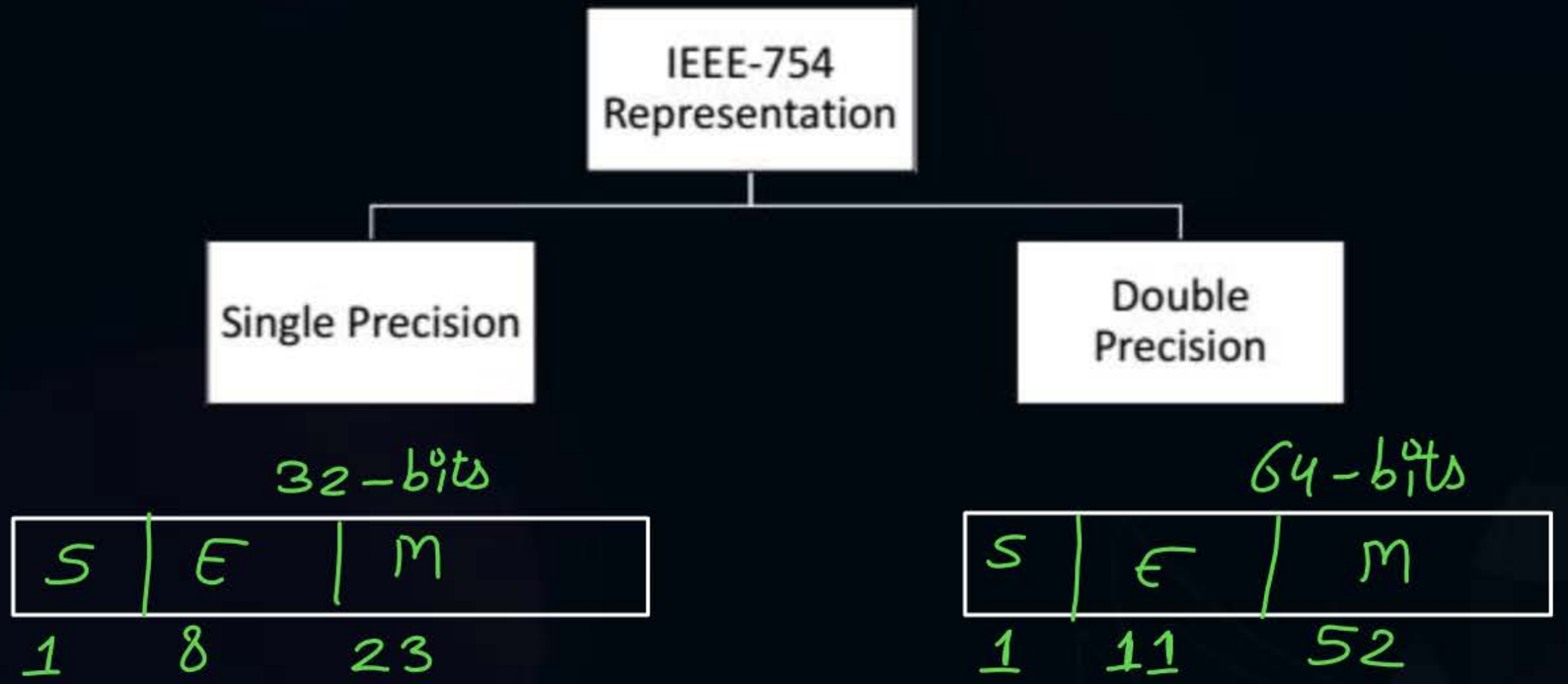
→ we can not store zero. (Implicit ⇒ 1. m, Explicit ⇒ 0.1)

→ we can not store small values around zero because of underflow.



Topic : IEEE-754 Floating Point Representation





$$E = \begin{cases} 00\ldots0 \\ \text{or} \\ 11\ldots1 \end{cases}$$

⇓

represent special number



Topic : IEEE-754 Floating Point Representation

S	E	M	Number
0	00.....0	0.....0	+0
1	00.....0	0.....0	−0
0	11.....1	0.....0	+∞
1	11.....1	0.....0	−∞
0 or 1	11.....1	M ≠ 0.....0	N.A.N. (Not a number)
0 or 1	00.....0	M ≠ 00.....0	Denormalized number
0 or 1	E ≠ 0.....0 and E ≠ 1.....1	Anything → 00.....0 to 11.....1	Implicit normalized number (normal number)



2 mins Summary

Topic RISC vs CISC

Topic Floating-Point Numbers

Topic Biased Exponent

Topic Number Range

Topic IEEE-754 Floating Point Representation



Happy Learning

THANK - YOU